Homework 8

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Problème 1

$T_c = 500$ ps

$t_s = 65$ ps

$t_{pcq} = 50$ ps

$\Rightarrow t_{pd} \leq T_c - (t$

$\Rightarrow t_{pd} \leq 500 - (65 + 50)$

$t_{pd} \leq 385$ ps

hence max delay is 385 ps
(max propagation delay)

Problème 2

$T_c = 500$ ps

$t_s = 65$ ps

$t_{pcq} = 50$ ps

since we add a clock skew the formula is:

$\Rightarrow t_{pd} \leq T_c - (t_{pcq} + t_{setup} + t_{skew})$

$\Rightarrow t_{pd} \leq 500 - (65 + 50 + t_{skew})$

$t_{skew} = 50$ ps

$\Rightarrow t_{pd} \leq 500 - (65 + 100)$

$\Rightarrow t_{pd} \leq 335$ ps

hence max delay for a second flip flop is 335 ps

Problème 3

$t_{ccq} = 35ps$

$t_{hold} = 30ps$

$\Rightarrow t_{cd} \geq t_{hold} - t_{ccq}$

$\Rightarrow t_{cd} \geq 30 - 35$

$\Rightarrow t_{cd} \geq -5ps$

Since minimum contamination delay is negative it is not physically meaningfull, $t_{cd} \geq 0ps$

The minimum contamination delay is 0 ps.

Problème 4

$t_{ccq} = 35ps$

$t_{hold} = 30ps$

$t_{skew} = 50ps$

since we have a clock skew the formula is:

$\Rightarrow t_{cd} \geq t_{hold} - t_{ccq} + t_{skew}$

$t_{cd} \geq -5 + 50$

$t_{cd} \geq 45 ps$

The minimum contamination delay for a skewed flipflop is 45 ps.